DISEASE PREDICTION USING MACHINE LEARNING ALGORITHMS

ABSTRACT:

In today’s world, technology has become an integral part of improving the healthcare sector. One of the most impactful technologies being utilized today is machine learning, which assists in the analysis of medical information to predict diseases in their early stages. Disease prediction is beneficial to the patient and improves on the services that the healthcare organizations deliver. Advanced data analysis techniques allows physicians to be more accurate with the diagnosis which can be life saving.

This project is aimed at creating a system that helps physicians to determine the illness a patient suffers from more accurately. A dataset of 4,920 patient records each diagnosed with 41 different diseases was used for the study. Analyses focused on 95 of the 132 symptoms that were selected as key symptoms associated with different diseases over a variety of complex diseases. The development of the system was done using machine learning algorithms which are capable of diagnosing diseases by recognizing patterns in the data.

The project implements three well known machine learning models to achieve this purpose: Decision Tree Classifier, Random Forest Classifier, and Naïve Bayes Classifier. These algorithms help in determining the relations within the symptoms and diseases ensuring that physicians diagnose their patients accurately and faster. The outcome of these algorithms is analyzed and the algorithm that yields the best results in precise disease prediction is selected.

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